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United States Patent	12415226
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Liu; Shuang et al.

Methods and apparatus for pulse arc starting phase for welding

Abstract

Provided is a system for providing pulsed arc starting phase, where the system comprises power conversion circuitry configured to convert input power to welding-type power and output the welding-type power, and control circuitry configured to control the power conversion circuitry to output the welding-type power. The control circuitry is configured to control the power conversion circuitry to output a plurality of welding current pulses during at least a portion of one or both of a run-in period or a ramp period for wire feeding of a welding wire ends, where each of the plurality of welding current pulses is associated with a respective pulse period and a respective pulse duty cycle.

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Appl. No.: 17/002925

Filed: August 26, 2020

Prior Publication Data

Document Identifier	Publication Date
US 20210060679 A1	Mar. 04, 2021

Related U.S. Application Data

us-provisional-application US 62894177 20190830

Publication Classification

Int. Cl.: B23K9/09 (20060101); B23K9/073 (20060101); B23K9/095 (20060101); B23K9/173 (20060101)

U.S. Cl.:

CPC B23K9/091 (20130101); B23K9/073 (20130101); B23K9/0953 (20130101); B23K9/0956 (20130101); B23K9/173 (20130101);

Field of Classification Search

CPC: B23K (9/10); B23K (9/1031); B23K (9/067); B23K (9/073); B23K (9/091); B23K (9/092); B23K (9/0953); B23K (9/0956); B23K (9/095); B23K (9/1087); B23K (9/1006); B23K (9/125); B23K (9/124); B23K (9/133); B23K (9/1336); B23K (9/1333); B23K (9/173); B23K (9/09-093); B23K (9/06-0671)

USPC: 219/121.11; 219/121.13; 219/130.5-13; 219/132; 219/137PS; 219/137.2

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Background/Summary

RELATED APPLICATIONS (1) This patent claims priority to U.S. Provisional Patent Application Ser. No. 62/894,177, filed Aug. 30, 2019, entitled “Method and Apparatus for Pulse Arc Starting Phase for Welding.” The entirety of U.S. Provisional Patent Application Ser. No. 62/894,177 is incorporated herein by reference.

BACKGROUND

- (1) The present disclosure relates to providing an arc for welding, and more particularly, to a methods and apparatus for pulse arc starting phase for welding.
- (2) When welding is first started, providing welding arc with a welding wire, such as, for example, a metal core wire, using a constant voltage power source may cause problems where the welding wire can break and/or cause sputtering of melted metal over the workpiece. A broken welding wire will result in additional time to start a welding arc, and spattered metal may need to be removed adding additional work to finishing a welding job.
- (3) Limitations and disadvantages of conventional systems for heating a workpiece for welding will become apparent to one of skill in the art, through comparison of such approaches with some aspects of the present method and system set forth in the remainder of this disclosure with reference to the drawings.

SUMMARY

- (4) Methods and systems are provided for pulse arc starting phase for welding, substantially as illustrated by and described in connection with at least one of the figures, as set forth more completely in the claims.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) These and/or other aspects will become apparent and more readily appreciated from the following description of some various aspects of examples of the disclosure, taken in conjunction with the accompanying drawings.
- (2) FIG. 1 is a high-level block diagram of system for providing welding current to a welding torch, in accordance with aspects of the disclosure.
- (3) FIG. 2 shows a block diagram of an example control circuitry, in accordance with aspects of the disclosure.
- (4) FIG. 3 shows a flow diagram representative of example machine readable instructions **300**, which may be executed by the example control circuitry **110** and/or **200** of FIGS. 1 and 2 to implement the arc start pulse control phase for pulse arc starting phase for the system **100** of FIG. 1.
- (5) FIG. 4 shows graphs that illustrate example pulse control signals output by the control circuitry **110** and/or **200** of FIGS. 1 and 2 and example wire feed speed for the system **100** in FIG. 1.
- (6) The figures are not necessarily to scale. Where appropriate, similar or identical reference numbers are used to refer to similar or identical components.

DETAILED DESCRIPTION

- (7) Various examples of the disclosure improve a starting phase for welding to alleviate broken welding wires and/or sputtering of molten metal onto a workpiece. For example, when welding is first initiated, controlled pulses of welding current may be provided to a welding wire to alleviate overheating that can result in breaking the welding wire and/or also creating excess spatter. Conventional systems may break the welding wire at the location at which current is transferred to the welding wire (e.g., the contact point between the welding wire and a contact tip).

(8) Keeping the welding wire from breaking at undesired places can help speed up the welding process because the welding wire does not need to be fed to the workpiece to make up for the broken welding wire. Additionally, reducing spatter caused by excess energy in the welding wire reduces grinding and/or other non-welding work to clean up the spatter.

(9) FIG. 1 is a high-level block diagram of system for providing welding current to a welding torch, in accordance with aspects of the disclosure. Referring to FIG. 1, there is shown an example system **100** comprising control circuitry **110**, a power source **120**, power conversion circuitry **130**, a wire feed motor **140**, and a gas supply **150**. The control circuitry **110** may also comprise sensors **160**. There is also shown a welding torch **170** that receives welding-type power from the system **100**, as well as the welding wire fed by the wire feed motor **140** and gas from the gas supply **150**. The control circuitry **110** is shown in more detail in FIG. 2.

(10) The control circuitry **110** is configured to output control signals to the power conversion circuitry **130** so that input power to the power conversion circuitry **130** can be converted to welding-type power, and the welding-type power can be output for use by the welding torch **170**. The power conversion circuitry **130** converts the input power to output power pulses where at least one of the power pulses is controlled for pulse period and pulse duty cycle. The control circuitry **110** may control, for example, one or both of the current and voltage. For brevity, in the following examples the current will be described as being controlled as current pulses.

(11) The current pulses may be output by the power conversion circuitry **130** during and after a run-in period or only after the run-in period. As used herein, run-in period is defined as the period of time when a welding wire feeder is turned on and feeds the welding wire at an initial wire feed rate. The run-in period may end, for example, when there is a first strike of the welding wire to the workpiece. An example run-in period is represented in FIG. 4 run-in as ending when the welding wire feeder starts to ramp up the wire feed rate.

(12) The control circuitry **110** is configured to control the current pulses output by the power conversion circuitry **130** by controlling a pulse period and a pulse duty cycle. Accordingly, each current pulse may have a specific period and duty cycle associated with the current pulse, and the period and/or duty cycle may be different from pulse to pulse. However, several current pulses can have the same period and/or duty cycle. The duty cycle and/or the period for the current pulses may be pre-determined or calculated in real time by, for example, the control circuitry. Calculating in real-time may entail, for example, calculating the pulse duty cycle and/or the pulse period in time to output the current pulse without delay. If pre-determined, the control circuitry **110** can read the pulse information such as the pulse period and pulse duty cycle values from, for example, memory, where the values may be stored in a lookup table. For example, the pulse information for a next pulse can be retrieved from memory, or calculated if the information is not retrieved from memory, prior to the start of the next pulse so that the next pulse can be controlled appropriately.

(13) The pulse period and the pulse duty cycle may be based on, for example, a type of welding wire and a shield gas to be used in the welding process. A pulse period may be considered to be from, for example, a rising edge of a pulse to the rising edge of the next pulse (time period T_a - T_b in FIG. 4), or from the falling edge of a pulse to a falling edge of the next pulse (time period T_y - T_z in FIG. 4). Accordingly, pulse information can be retrieved or generated during a previous pulse period. For example, the pulse information may be provided during a non-active portion of a pulse (for example, time period T_y - T_b in FIG. 4), an active portion of a pulse (for example, time period T_a - T_y in FIG. 4), or at any period during a pulse (for example, time period T_a - T_b in FIG. 4) to allow an active portion of the next pulse (for example, time period T_b - T_z) to be shaped appropriately according to the pulse information comprising the period and the duty cycle. It should be noted that while a pulse is shown as an active period followed by an inactive period, a pulse may also be described as an inactive period followed by an active period. Furthermore, while an example was given of providing pulse information during the previous pulse period, an example of the disclosure may provide the pulse information before the previous pulse period.

(14) In an example of the disclosure, the pulse information, for one or more welding processes that each have a unique welding wire type and welding gas combination, may be downloaded from another electronic device either wirelessly or via a wired connection. The wireless or wired download may be from, for example, another server that generates and/or stores welding information and downloads the appropriate information to the system **100** as needed. Additionally, a wired download may be from a memory device including, for example, the memory **220** or a flash drive that can plug into a USB port. The pulse information may have been downloaded to the memory device by a server, for example. The pulse information may be read for one or more pulses from the memory device.

(15) In an example of the disclosure, the pulse information for one or more pulses may be requested via wire or wirelessly using the communication interface **230**. The pulse information may be provided by, for example, an electronic device configured to communicate via one or both of wired or wireless communication.

(16) Some examples also provide at least one pulse during the run-in period for a welding process. For example, FIG. **4** shows an active portion of a pulse **402** during the time period **T1-T3**. While time **T3** is shown to coincide substantially with the end of the run-in period **P1** (FIG. **4**), various examples of the disclosure may have the active portion of a pulse **402** end prior to the end of the run-in period **P1**, or after the end of the run-in period **P1**. Also, the start time **T1** of the pulse **402** is shown as an example to be after the time **T0**, however, the start time **T1** may be any time at **T0** or after **T0**. It should be noted that time **T0** may be the time when power is available to generate the pulses **402**, **404**, and **406** shown in FIG. **4**.

(17) Additionally, while a single pulse **402** has been shown for the sake of brevity, various examples can have multiple pulses **402** during the run-in period **P1**. Therefore, an active period of the first of the pulses **402** can start at time **T0** or after time **T0**, and an active period of the last of the pulses **402** can end before time **T3**, at time **T3** or after time **T3**. Accordingly, there may be multiple pulses **402** comprising one or more pulses **402a** and one or more pulses **402b**. The pulses **402a** and **402b** may be of constant period and duty cycle, or the period and/or duty cycles may vary.

(18) Similarly as described above, the pulse information for the next pulse may be retrieved or generated prior to the start of a present pulse period. It should also be noted that the pulse information for a pulse may be provided after the active portion of that pulse starts (for example, after time **Ta** for a first pulse **404**) as long as the pulse period and duty cycle for a first of the pulses **404** can be controlled appropriately.

(19) Additionally, once the welding wire is at a steady wire feed speed at time **T4**, for example, the current pulses may be kept constant with respect to the pulse period and the pulse duty cycle. The steady wire feed speed may be known for a welding process that the system **100** is configured for, or the steady wire feed speed may be a commanded wire feed speed that is entered by a user via, for example, the input devices **242** or received via the communication interface **230** in FIG. **2**.

(20) The control circuitry **110** may continue to receive feedback regarding the wire feed speed during the welding process, and determine/retrieve new pulse information based on the wire feed speed. The wire feed speed may be provided by any of several methods such as, for example, a predictive speed filter, monitoring a motor speed of a wire feed motor **140**, and/or monitor a travel speed of the welding wire. These methods for feeding back wire feed speed information may be determined by using any appropriate technology presently known or to be developed in the future. For example, the predictive speed filter may be a Kalman filter where the instructions to perform the filtering may be in the memory **220** (FIG. **2**).

(21) The feedback information may be provided by, for example, sensors **160** that sense the speed of the wire motion or motor speed, the control circuitry **110** that controls the wire feed motor **140**, etc. These sensors **160** may be, for example, in the system **100**. Various sensors may also be in the welding torch **170** (sensors in the welding torch **170** are not shown) or in another module (not

shown).

(22) Additionally, other examples of the system **100** may have the control circuitry **110** control a current level of the output pulses to add an additional level of control to the system **100**.

(23) While an example of the system **100** is shown as having the control circuitry **110**, the power source **120**, and the power conversion circuitry **130** in separate blocks, another example of the disclosure may have a different grouping of the control circuitry **110**, the power source **120**, the power conversion circuitry **130**. For example, the power source **120** and the power conversion circuitry **130** may be in one block, or the control circuitry **110**, the power source **120**, and the power conversion circuitry **130** may be in one block. Similarly, the wire feed motor **140**, the gas supply **160**, and the sensors **170** may be grouped differently than shown in FIG. 1. Accordingly, it can be seen that various examples of the disclosure may configure the system **100** differently depending on the design criteria for the system **100**.

(24) Referring to FIG. 2, there is shown an example control circuitry **200** that may be used with various examples of the disclosure, and may be similar to the control circuitry **110** in FIG. 1. The control circuitry **200** may comprise, for example, a processor **210**, memory **220**, a communication interface **230**, and an IO interface **240**. The processor **210** may comprise, for example, one or more of processors (CPUs, GPUs, etc.), controllers, system on chips, ASICs, etc.

(25) The memory **220** may include non-volatile memory **226** and volatile memory **228**. The storage described for holding local data may be part of the memory **220** or comprise separate memory. The operating system **222**, applications **224**, and/or data such as a look up table, may be stored in, for example, the non-volatile memory **226**, and may be copied to volatile memory **228** for execution/access by the processor **210**. Various aspects of the disclosure may use different memory architectures that are design and/or implementation dependent. For example, some aspects of the disclosure may have the operating system **222** and applications **224** in the non-volatile memory **226** executed at least in part from the non-volatile memory **226**.

(26) The communication interface **230** may allow the control circuitry **200** to communicate with other devices via, for example, a wired protocol such as USB, Ethernet, Firewire, etc., or a wireless protocol such as Bluetooth, Near Field Communication (NFC), Wi-Fi, etc. The wired or wireless protocol may also be, for example, a proprietary protocol. The various types of radios for communication may be referred to as a transceiver for the sake of simplicity. The communication may be, for example, with various sensors and/or devices that can relay sensor data. The communication interface **230** may also be used to communicate with other networks such as local networks, cellular networks, etc. The IO module **240** may allow various devices to plug into, for example, a USB port. For example, a keyboard and/or a mouse may plug into their respective USB ports, or communicate wirelessly to a USB dongle that allows wireless communication with, for example, a wireless keyboard and/or mouse. A USB port may also be used to download and/or upload data from/to a memory device such as, for example, FLASH drive.

(27) The control circuitry **200** may also comprise the IO module **240** for communication with a user via the input devices **242** and output information to be displayed on output devices **244**. The input devices **242** may comprise, for example, switches, slide switches, membrane switches, buttons, touch sensitive screen, which may be a part of a display, a microphone, etc. The touch sensitive screen (touchscreen) may have soft buttons, switches, slide switches, keyboard, etc. that emulate their physical counterparts. The input devices **242** may also comprise, for example, a keyboard, a mouse, a trackball, etc., as well as various sensors, cameras, etc. The input devices **242** may additionally comprise, for example, bar code readers and/or other type of scanners that may be used to identify an object. The output devices **244** may comprise, for example, display(s), speaker(s), LED(s), vibration motor(s), etc. Some devices such as a touchscreen are able to provide both input and output functions of the IO module **240**.

(28) The processor **210** may operate using different architectures in different examples of the disclosure. For example, the processor **210** may use the memory **220** to store instructions to

execute, or the processor **210** may have its own memory (not shown) for its instructions.

(29) Various examples may use other architectures where the different functionalities may be grouped differently. For example, the functionalities may be in different integrated circuit chips, or different devices may be combined. In a non-limiting example, the IO module **240** and the communication interface **230** may be combined together. Additionally, the control circuitry **200** may refer logically to various physical devices. For example, one or more of the output devices **244** may be part of a different integrated circuit or on a different printed circuit board than one or more of the input devices **242**.

(30) FIG. **3** shows a flow diagram representative of example machine readable instructions **300**, which may be executed by the example control circuitry **110**, **200** of FIGS. **1** and/or **2** to implement the arc start pulse control phase for pulse arc starting phase for the system **100** of FIG. **1**. For example, the instructions **300**, which may be in an application **224**, may be executed by the processor **210** to control the system **100** to output current pulses for an arc starting phase for welding.

(31) In block **302**, the system **100** may wait for a start request for a welding procedure. The start request may be received via, for example, the input devices **242** of FIG. **2** and/or the start request may occur when the system **100** is turned on. In block **304**, a controller such as, for example, the control circuitry **110** or **200** may start the wire feed motor **140**. The wire feed motor **140** may then proceed to feed the wire at a run-in speed as shown in FIG. **4**. In block **306**, the current pulses are initiated to start an arc with the welding torch **170** for the welding process. Each current pulse is controlled for its period and its duty cycle to appropriately heat the welding wire without causing it to break or sputter.

(32) In block **308**, wire feed speed information is received from, for example, a predictive speed filter such as a Kalman filter, or from a feedback mechanism that reports the wire feed speed or information that can be processed to determine the wire feed speed. In some cases, the commanded wire feed speed may be used rather than the wire feed speed information.

(33) In block **310**, the wire feed speed is used to control at least the next current pulse, depending on the feedback rate and processing of the wire feed speed information. In block **312**, it may be determined whether the welding process is finished. For example, when the welding torch trigger is released, the welding process may be assumed to be finished for the time being and the wire feed motor **140** may be turned off. The instructions **300** may then proceed to block **302** to wait for a start request.

(34) While an example flow diagram is shown in FIG. **3** for illustrative purposes, it should be understood that various other flow diagrams may also describe other examples of the disclosure. For example, in another example of the disclosure, the blocks **304** and **306** may be performed in series rather than in parallel as shown in FIG. **3**.

(35) FIG. **4** shows graphs that illustrate example pulse control signals output by the control circuitry **110** and/or **200** of FIGS. **1** and **2** and example wire feed speed for the system **100** in FIG. **1**. Graph **400** shows a graph of I.sub.CMD versus time, where I.sub.CMD is a control signal from the control circuitry **110** used by the power conversion circuitry **130** to output current pulses for use in welding. Graph **410** shows a graph of wire feed speed in inches/min (IPM) versus time.

(36) In the example shown, graph **400** shows a single active pulse **402** (time period T1-T3) during the run-in period P1 (time period T0-T3), and a plurality of pulses **404** and **406** after the run-in period P1. However, it should be understood that the control signal I.sub.CMD can start (be active) at any time prior to the run-in period P1, as well as during the run-in period P1. That is, while FIG. **4** discloses an example of the control signal I.sub.CMD during the time period T1-T3 (pulse **402**), other examples may have the control signal I.sub.CMD start at time T0 or any time after time T0. The run-in period P1 and the ramp period P3 (time period T3-T4) are shown in the graph **410**. The period after T4 in graph **410** is the period when the wire feed speed has reached a desired speed for the welding process.

(37) Furthermore, while a single pulse **402** is shown as an example pulse that starts after time **T0** during the run-in period **P1** in FIG. **4**, other aspects of the disclosure may have a single pulse **402** that starts at time **T0**. Still other aspects of the disclosure may have two or more pulses **402** during the run-in period **P1**, where the first pulse starts at either time **T0** or after time **T0**. Furthermore, there may be multiple pulses during the run-in period **P1** that may comprise a pulse **402a** and one or more pulses **402b**, where the various pulses may have similar or different active and inactive times. Accordingly, each pulse may be controlled as to its active and inactive times.

(38) Similarly, while three pulses **404** are shown as an example during the ramp period **P3**, various aspects of the disclosure need not be so limited. Accordingly, there may be a different number of pulses **404** in a different aspect of the disclosure. The pulses **404** are also shown to have different active and inactive times, however, various aspects of the disclosure need not be so limited. Accordingly, two or more pulses **404** may have similar active and/or inactive times. Accordingly, in an aspect of the disclosure, the pulses **404** may be fixed rather than controlled in coordination with the wire feed speed.

(39) Similarly, while the pulses **406** are shown to have similar active and inactive times, various aspects of the disclosure need not be so limited. Accordingly, one or more of the pulses **406** may be controlled to have different active and/or inactive times as well as different pulse periods.

(40) Additionally, various aspects of the disclosure may provide for an “initiate period **P2**” that can occur, for example, during a portion of the pulse **402**. The initiate period **P2** may be at the beginning of the pulse **402** during the time period **T1-T2** as shown, or the initiate period **P2** may start after the beginning of the pulse **402**. During the initiate period **P2**, there may be provided, for example, constant voltage, constant current, constant energy, pulsed voltage, pulsed current, and/or pulsed energy.

(41) It should be noted that while multiple pulses **402b** are provided after the initiate period **P2** (time period **T1-T2**), various examples of the disclosure need not be so limited. For example, the pulses **402b** may start before time **T2** or after time **T2**.

(42) Accordingly, it can be seen that the present disclosure provides for a system for providing pulsed arc starting phase, comprising power conversion circuitry configured to convert input power to welding-type power, and to output the welding-type power. The system also comprises control circuitry configured to control the power conversion circuitry to output the welding-type power by controlling the power conversion circuitry to output a plurality of welding current pulses during at least a portion of one or both of a run-in period or a ramp period for wire feeding of a welding wire, where each of the plurality of welding current pulses is associated with a respective pulse period and a respective pulse duty cycle.

(43) One or more of the respective pulse period and the respective pulse duty cycle may be based on, for example, a welding wire type and a shield gas. The control circuitry may be configured to perform one or both of retrieve a pre-determined pulse period and a pre-determined pulse duty cycle from memory, or determine the respective pulse period and the respective pulse duty cycle in real time. The pre-determined pulse period and the pre-determined pulse duty cycle for a subsequent welding current pulse can be retrieved by the control circuitry during, for example, the pulse period of a present welding current pulse. The respective pre-determined pulse period and the respective pre-determined pulse duty cycle can also be received from an external electronic device via one or both of wireless communication and wired communication, and the received pre-determined pulse period and the received respective pre-determined pulse duty cycle can be stored in memory.

(44) The control circuitry is configured to provide one or more pulses during at least a portion of the run-in period, where at least one of the one or more pulses comprises one or more of constant voltage, constant current, or constant energy.

(45) The control circuitry is also configured to provide at least one first welding current pulse during the run-in period, and end a last of the at least one first welding current pulse substantially

when the run-in period ends.

(46) The output of the plurality of welding current pulses may be correlated to a wire feed speed, where the wire feed speed can be provided by a predictive speed filter, such as, for example, a Kalman filter. The wire feed speed may also be provided by, for example, a commanded wire feed speed. The control circuitry may also be configured to determine the wire feed speed of the welding wire by monitoring a motor speed of a wire feed motor and/or a travel speed of the welding wire.

(47) The disclosure can also be seen to provide for a method for providing pulsed arc starting phase, comprising converting, during at least a portion of one or both of a run-in period or a ramp period for feeding a welding wire, an input welding current to output a plurality of welding current pulses, where each of the plurality of welding current pulses is associated with a respective pulse period and a respective pulse duty cycle. The respective pulse period and the respective pulse duty cycle may be based on, for example, a welding wire type and a shield gas. The method may also comprise one of determining the respective pulse period and the respective pulse duty cycle in real time or retrieving pre-determined pulse periods and pre-determined pulse duty cycles from memory.

(48) The pre-determined pulse period and the pre-determined pulse duty cycle for a subsequent welding current pulse may be retrieved, for example, during the pulse period of a present welding current pulse.

(49) The method may further comprise receiving the pre-determined pulse periods and the pre-determined pulse duty cycles from an external electronic device via one or both of wireless communication and wired communication, and storing the received pre-determined pulse periods and the pre-determined pulse duty cycles in the memory.

(50) The output of the plurality of welding current pulses may be correlated to a wire feed speed provided by using one of a predictive speed filter, commanded wire feed speed, and a speed determined by monitoring one or both of a motor speed of a wire feed motor and a travel speed of the welding wire. The predictive speed filter may comprise, for example, a Kalman filter.

(51) The method may also provide for providing one or more pulses during at least a portion of the run-in period, where at least one of the one or more pulses comprises one or more of constant voltage, constant current, or constant energy.

(52) The present disclosure may also provide for a system for providing pulsed arc starting phase, comprising power conversion circuitry configured to convert input power to welding-type power. The power conversion circuitry may also be configured to output the welding-type power. The system may comprise control circuitry configured to control the power conversion circuitry to output the welding-type power by controlling the power conversion circuitry to output at least one first welding current pulse during a run-in period for wire feeding of a welding wire, including respective first pulse periods and respective first pulse duty cycles of the at least one first welding current pulse, and controlling the power conversion circuitry to output a plurality of second welding current pulses after the run-in period, including respective second pulse periods and respective second pulse duty cycles of the plurality of second welding current pulses, wherein the output of the welding-type power is correlated to a wire feed speed.

(53) The control circuitry may also be configured to provide one or more pulses during at least a portion of the run-in period, where at least one of the one or more pulses comprises one or more of constant voltage, constant current, or constant energy.

(54) The present methods and systems may be realized in hardware, software, and/or a combination of hardware and software. The present methods and/or systems may be realized in a centralized fashion in at least one computing system, or in a distributed fashion where different elements are spread across several interconnected computing systems. Any kind of computing system or other apparatus adapted for carrying out the methods described herein is suited. A typical combination of hardware and software may include a general-purpose computing system with a program or other

code that, when being loaded and executed, controls the computing system such that it carries out the methods described herein. Another typical implementation may comprise one or more application specific integrated circuit or chip. Some implementations may comprise a non-transitory machine-readable (e.g., computer readable) medium (e.g., FLASH memory, optical disk, magnetic storage disk, or the like) having stored thereon one or more lines of code executable by a machine, thereby causing the machine to perform processes as described herein. As used herein, the term “non-transitory machine-readable medium” is defined to include all types of machine readable storage media and to exclude propagating signals.

(55) As utilized herein the terms “circuits” and “circuitry” refer to physical electrical and/or electronic components (i.e. hardware) and any software and/or firmware (“code”) which may configure the hardware, be executed by the hardware, and or otherwise be associated with the hardware. As used herein, for example, a particular processor and memory may comprise a first “circuit” when executing a first one or more lines of code and may comprise a second “circuit” when executing a second one or more lines of code. As utilized herein, “and/or” means any one or more of the items in the list joined by “and/or.” As an example, “x and/or y” means any element of the three-element set {(x), (y), (x, y)}. In other words, “x and/or y” means “one or both of x and y”. As another example, “x, y, and/or z” means any element of the seven-element set {(x), (y), (z), (x, y), (x, z), (y, z), (x, y, z)}. In other words, “x, y and/or z” means “one or more of x, y and z”. As utilized herein, the term “exemplary” means serving as a non-limiting example, instance, or illustration. As utilized herein, the terms “e.g.” and “for example” set off lists of one or more non-limiting examples, instances, or illustrations. As utilized herein, circuitry is “operable” to perform a function whenever the circuitry comprises the necessary hardware and code (if any is necessary) to perform the function, regardless of whether performance of the function is disabled or not enabled (e.g., by a user-configurable setting, factory trim, etc.).

(56) As utilized herein, “and/or” means any one or more of the items in the list joined by “and/or.” As an example, “x and/or y” means any element of the three-element set {(x), (y), (x, y)}. In other words, “x and/or y” means “one or both of x and y.” As another example, “x, y, and/or z” means any element of the seven-element set {(x), (y), (z), (x, y), (x, z), (y, z), (x, y, z)}. In other words, “x, y and/or z” means “one or more of x, y and z.” As utilized herein, “one of x or y” or “one of x and y” is equivalent to any element of the set {(x), (y)}.

(57) As utilized herein, the term “exemplary” means serving as a non-limiting example, instance, or illustration. As utilized herein, the terms “e.g.” and “for example” set off lists of one or more non-limiting examples, instances, or illustrations. As utilized herein, circuitry is “operable” to perform a function whenever the circuitry comprises the necessary hardware and code (if any is necessary) to perform the function, regardless of whether performance of the function is disabled or not enabled (e.g., by a user-configurable setting, factory trim, etc.).

(58) While the present method and/or system has been described with reference to certain examples and/or aspects, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted without departing from the scope of the present method and/or system. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the present disclosure without departing from its scope. Therefore, the present method and/or system are not limited to the particular examples disclosed. Instead, the present method and/or system will include all implementations falling within the scope of the appended claims, both literally and under the doctrine of equivalents.

Claims

1. A system for providing a pulsed arc starting phase of a welding process, the system comprising: power conversion circuitry configured to convert input power to welding-type power, and to output the welding-type power; and control circuitry configured to control the power conversion circuitry

to output the welding-type power by: controlling the power conversion circuitry to output a plurality of welding current pulses during at least a portion of one or both of a run-in period or a ramp period for wire feeding of a welding wire, wherein each of the plurality of welding current pulses is associated with a respective pulse period and a respective pulse duty cycle, wherein the power conversion circuitry is configured to correlate the output of the plurality of welding current pulses to a wire feed speed and to receive feedback regarding the wire feed speed during the welding process to determine new pulse information, wherein the new pulse information is generated from a first pulse of the plurality of welding current pulses and used to correlate a second pulse of the plurality of welding current pulses to the wire feed speed, and wherein the second pulse immediately follows the first pulse.

2. The system of claim 1, wherein one or more of the respective pulse period and the respective pulse duty cycle are based on a welding wire type and a shield gas.

3. The system of claim 1, wherein the control circuitry is configured to perform one or both of: retrieve a pre-determined pulse period and a pre-determined pulse duty cycle from memory, or determine the respective pulse period and the respective pulse duty cycle in real time.

4. The system of claim 3, wherein the control circuitry is configured to retrieve the pre-determined pulse period and the pre-determined pulse duty cycle for a subsequent welding current pulse during the pulse period of a present welding current pulse.

5. The system of claim 3, wherein the respective pre-determined pulse period and the respective pre-determined pulse duty cycle are received via one or both of wireless communication and wired communication, and stored in the memory.

6. The system of claim 1, wherein the control circuitry is configured to provide one or more pulses during at least a portion of the run-in period, wherein at least one of the one or more pulses comprises one or more of constant voltage, constant current, or constant energy.

7. The system of claim 1, wherein the control circuitry is configured to: provide at least one first welding current pulse during the run-in period, and end a last of the at least one first welding current pulse when the run-in period ends.

8. The system of claim 1, wherein the output of the plurality of welding current pulses is correlated to the wire feed speed.

9. The system of claim 8, wherein the wire feed speed is provided by one or more of: a predictive speed filter; a commanded wire feed speed, or a determined wire feed speed of the welding wire determined by the control circuitry by monitoring one or both of a motor speed of a wire feed motor and a travel speed of the welding wire.

10. The system of claim 8, wherein the wire feed speed is provided by a Kalman filter.

11. A system for providing a pulsed arc starting phase, the system comprising: power conversion circuitry configured to convert input power to welding-type power, and to output the welding-type power; and control circuitry configured to control the power conversion circuitry to output the welding-type power by controlling the power conversion circuitry to output: at least one welding current pulse during a run-in period for wire feeding of a welding wire; and a plurality of welding current pulses during a ramp period for wire feeding of the welding wire, wherein a duty cycle of the plurality of welding current pulses increases as a feed speed of the welding wire increases during the ramp period.

12. The system of claim 11, wherein each of the plurality of welding current pulses is associated with a respective pulse period and a respective pulse duty cycle.

13. The system of claim 12, wherein one or more of the respective pulse period and the respective pulse duty cycle are based on a welding wire type and a shield gas.

14. The system of claim 12, wherein the control circuitry is configured to perform one or both of: retrieve a pre-determined pulse period and a pre-determined pulse duty cycle from memory, or determine the respective pulse period and the respective pulse duty cycle in real time.

15. The system of claim 14, wherein the control circuitry is configured to retrieve the pre-

determined pulse period and the pre-determined pulse duty cycle for a subsequent welding current pulse during the pulse period of a present welding current pulse.

16. The system of claim 14, wherein the respective pre-determined pulse period and the respective pre-determined pulse duty cycle are received via one or both of wireless communication and wired communication, and stored in the memory.

17. The system of claim 11, wherein the at least one welding current pulse during the run-in period comprises one or more of constant voltage, constant current, or constant energy.

18. The system of claim 11, wherein the control circuitry is further configured to control the power conversion circuitry to output the welding-type power by controlling the power conversion circuitry to output a second plurality of welding current pulses after completion of the ramp period, wherein a duty cycle of the second plurality of welding current pulses is constant.

19. A system for providing a pulsed arc starting phase, the system comprising: power conversion circuitry configured to convert input power to welding-type power, and to output the welding-type power; and control circuitry configured to control the power conversion circuitry to output the welding-type power by controlling the power conversion circuitry to output: at least one welding current pulse during a run-in period for wire feeding of a welding wire; a first plurality of welding current pulses during a ramp period for wire feeding of the welding wire; and a second plurality of welding current pulses after completion of the ramp period, wherein a duty cycle of the first plurality of welding current pulses increases as a feed speed of the welding wire increases during the ramp period, and wherein a duty cycle of the second plurality of welding current pulses is constant.

20. The system of claim 19, wherein the duty cycle of the second plurality of welding current pulses is greater than or equal to the duty cycle of the first plurality of welding current pulses upon completion of the ramp period.
